

Classifying Parkinson's Disease from Voice Features Using Kernel SVMs

1. Problem and Dataset

Parkinson's disease is a neurodegenerative disorder that can affect motor control, including the muscles involved in speech and voice production (1). As a result, acoustic changes in the voice may provide useful information for distinguishing Parkinson's disease recordings from healthy-control recordings (1). This project aimed to evaluate whether biomedical voice features could be used to classify Parkinson's disease status using a nonlinear machine learning model. The research question was: Can a support vector machine with a radial basis function kernel distinguish Parkinson's disease recordings from healthy-control recordings using biomedical voice measurements?

I used the UCI Oxford Parkinson's Disease Detection Dataset, a public biomedical dataset designed for classification (2). Each row represents one voice recording, and the outcome variable is status, coded as 0 for healthy control and 1 for Parkinson's disease. The dataset included 195 voice recordings and 22 continuous biomedical voice features, with no missing values. The predictors included measures of vocal frequency, jitter, shimmer, noise-to-harmonics ratio, harmonics-to-noise ratio, RPDE, DFA, spread, D2, and PPE. These features capture different aspects of pitch, amplitude variation, noise, and nonlinear complexity in the voice signal.

An important feature of the dataset was class imbalance. There were 147 Parkinson's disease recordings and 48 healthy-control recordings, meaning approximately three-quarters of the observations belonged to the Parkinson's group. This imbalance is important because a model could achieve apparently high accuracy by favouring the majority class. For this reason, I evaluated model performance using not only accuracy, but also sensitivity, specificity, F1 score, and ROC-AUC. I also used class weighting during model training to reduce bias toward the majority Parkinson's class.

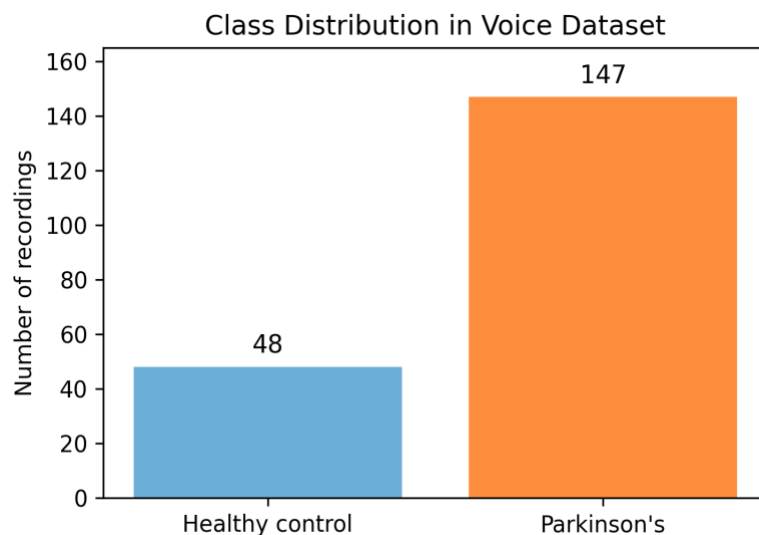


Figure 1. Class distribution in the UCI Parkinson’s voice dataset. The dataset contained 147 Parkinson’s disease recordings and 48 healthy-control recordings, showing clear class imbalance.

Table 1. Summary of the UCI Parkinson’s voice dataset.

Dataset feature	Value
Total recordings	195
Parkinson’s recordings	147
Healthy-control recordings	48
Predictors	22 continuous voice features
Outcome	Status: 0 = healthy control, 1 = Parkinson’s disease
Missing values	None

2. Machine Learning Algorithm: RBF-Kernel SVM

To classify Parkinson’s disease status, I used a support vector machine with a radial basis function kernel (3). A support vector machine is a supervised classification algorithm that aims to find a decision boundary separating two classes while maximizing the margin between them. The margin is the distance between the decision boundary and the closest training observations from each class. These closest observations are called support vectors, and they are important because they help determine the position of the decision boundary.

In this project, the two classes were healthy-control recordings and Parkinson’s disease recordings. A linear SVM would try to separate these groups using a straight boundary in the original feature space. However, biomedical voice features are unlikely to distinguish Parkinson’s disease from healthy controls in a perfectly linear way, because pitch variation, amplitude variation, noise measures, and nonlinear voice features may interact in complex ways. For this reason, I used a radial basis function kernel, which allows the SVM to learn a flexible nonlinear decision boundary.

The RBF kernel measures similarity between two observations based on their distance in feature space:

$$K(x_i, x_j) = \exp\left(-\gamma \left\|x_i - x_j\right\|^2\right)$$

Here, x_i and x_j represent two voice recordings, and $\left\|x_i - x_j\right\|^2$ is the squared distance between their feature values. If two recordings have similar acoustic profiles, this distance is small and the kernel value is high. If they are very different, the distance is larger and the kernel value becomes smaller. The parameter γ controls how quickly similarity decreases as recordings become farther apart. This allows the model to classify observations based on similarity without manually creating nonlinear interaction terms.

Because real biomedical data are rarely perfectly separable, I used a soft-margin SVM. A soft-margin SVM allows some observations to fall inside the margin or even on the wrong side of the decision boundary. These violations are represented by slack variables, usually written as ξ_i . The soft-margin SVM balances two goals: keeping the margin wide and penalizing classification errors. This trade-off can be written as minimizing:

$$\frac{1}{2} ||w||^2 + C \sum_i \xi_i$$

The first term encourages a wider and simpler margin, while the second term penalizes margin violations. The parameter C controls how strongly these violations are punished. A smaller C allows more violations and may produce a wider, simpler margin, while a larger C penalizes mistakes more heavily and may produce a tighter, more complex boundary. Therefore, both C and γ control model flexibility and must be tuned carefully to avoid underfitting or overfitting.

This algorithm is appropriate for this problem because RBF-kernel SVMs often work well on small-to-medium sized datasets with continuous numerical predictors, especially when the relationship between predictors and outcome may be nonlinear. The main assumption behind using an RBF-kernel SVM is that the kernel-based similarity measure is meaningful for the classification task. In this context, voice recordings that are close together in the standardized acoustic feature space are assumed to be more likely to share the same disease label. This is reasonable for this dataset because the predictors capture related aspects of voice impairment, including pitch variation, amplitude variation, noise, and nonlinear signal complexity. However, because SVMs rely on distances between observations, the features must be properly scaled. In addition, if C or γ are too large, the model may overfit to noise in the training data. For this reason, both hyperparameters were tuned using cross-validation rather than chosen manually.

3. Results and Discussion

Before fitting the model, all 22 voice features were standardized because RBF-kernel SVMs rely on distances between observations. The data were then split into training and test sets using a stratified 75/25 split, so that the proportion of Parkinson's disease and healthy-control recordings was similar in both sets. To address class imbalance, I used balanced class weights during training, which penalized errors on the smaller healthy-control class more strongly. The model was implemented in Python using scikit-learn, and the main SVM hyperparameters, C and γ , were tuned using grid search with five-fold cross-validation (4). The best-performing tuned model used $C = 100$ and $\gamma = 0.1$.

After hyperparameter tuning and class weighting, I evaluated different probability thresholds as a threshold sensitivity analysis. A threshold of 0.8 gave the best balance between sensitivity and specificity. At this threshold, the final model achieved a test ROC-AUC of 0.96 and an accuracy of 0.94. It correctly classified 36 of 37 Parkinson's recordings and 10 of 12 healthy-control recordings, corresponding to Parkinson's sensitivity of 0.97 and healthy-control specificity of 0.83. The macro F1 score was 0.91, suggesting strong performance across both classes despite the class imbalance.

Table 2. Test-set performance of the tuned RBF-kernel SVM at threshold 0.8.

Metric	Test performance
ROC-AUC	0.96
Accuracy	0.94
Sensitivity	0.97
Specificity	0.83
Macro F1	0.91

Although the overall accuracy was high, the confusion matrix was important because it showed performance separately for Parkinson's disease and healthy-control recordings, rather than allowing the majority Parkinson's class to dominate the interpretation. In the context of Parkinson's disease screening, the high sensitivity of 0.97 means that the model missed very few Parkinson's disease recordings in the test set. This matters because a screening model should ideally minimize false negatives, since missing a possible Parkinson's case could delay further clinical assessment. The specificity of 0.83 means that the model also correctly identified most healthy-control recordings, although some healthy recordings were still incorrectly classified as Parkinson's disease. This trade-off may be acceptable for a preliminary screening tool, where false positives could lead to additional assessment, but false negatives may be more clinically concerning. However, because this analysis used a small recording-level dataset, these results should be interpreted cautiously.

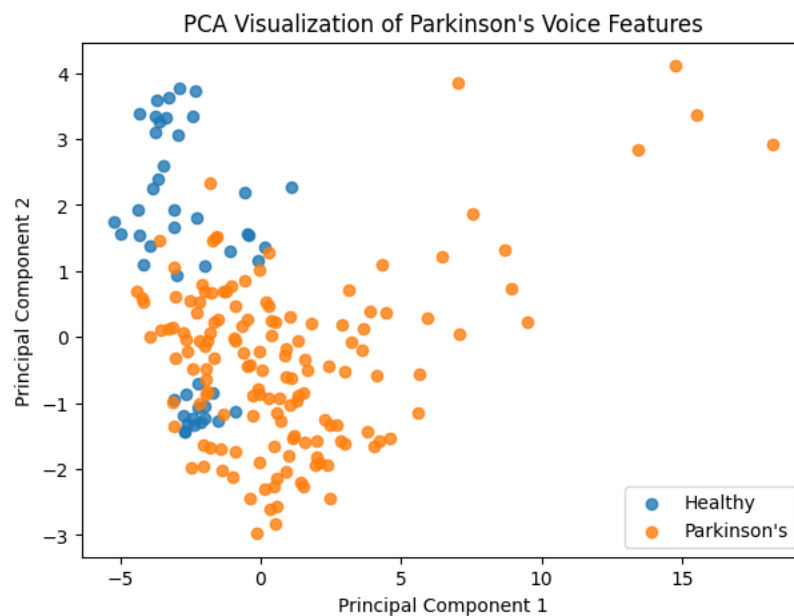


Figure 2. PCA visualization of Parkinson's voice features. The 22 biomedical voice features were projected onto the first two principal components for visualization. Healthy-control recordings clustered mainly toward the left and upper-left of the plot, while Parkinson's recordings were more widely distributed across the center and right side. The visible overlap between groups suggests that the classes are not perfectly separable in two dimensions, supporting the use of a nonlinear classifier such as an RBF-kernel SVM.

Although the PCA plot was used only for visualization and not for model training, the overlap between groups in two dimensions provides additional support for the choice of a nonlinear classifier. The RBF-kernel SVM used the full 22-dimensional standardized feature space rather than relying only on the two principal components shown in the plot.

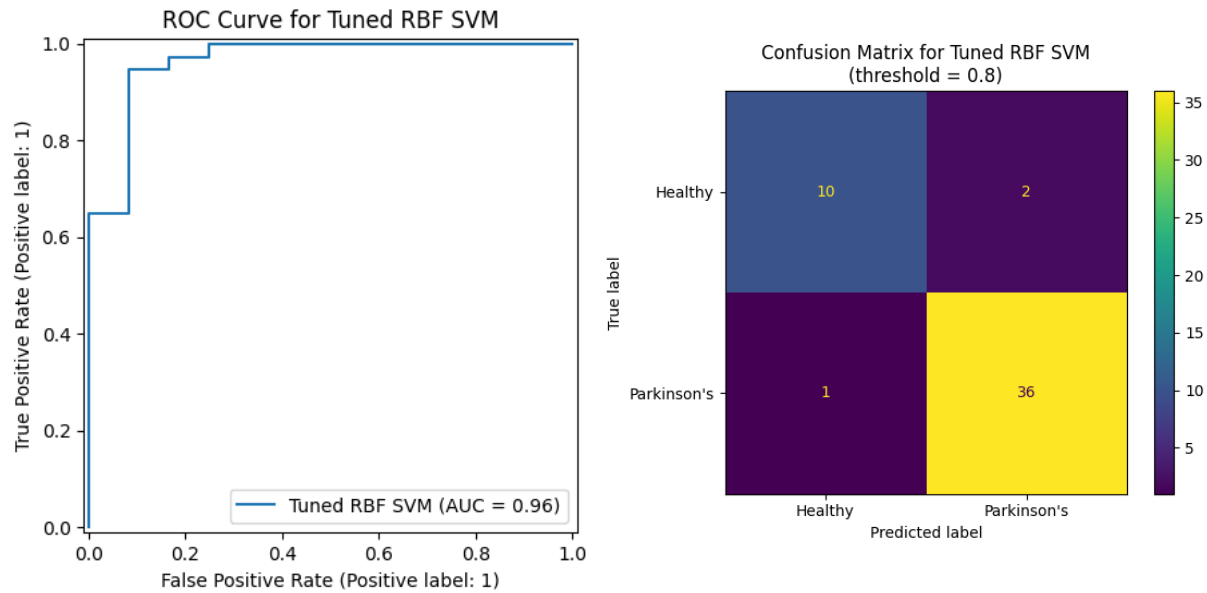


Figure 3. Performance of the tuned RBF-kernel SVM. The ROC curve showed strong discrimination between Parkinson’s disease and healthy-control recordings, with an AUC of 0.96. The confusion matrix shows performance at the selected probability threshold of 0.8, where the model correctly classified 36 of 37 Parkinson’s recordings and 10 of 12 healthy-control recordings.

These results suggest that biomedical voice features contain useful information for distinguishing Parkinson’s disease recordings from healthy controls. The high ROC-AUC indicates that the model was effective at ranking Parkinson’s recordings above healthy-control recordings across thresholds. The threshold sensitivity analysis was important because changing the classification threshold affected the balance between sensitivity and specificity. In this case, a threshold of 0.8 improved healthy-control classification while maintaining high sensitivity for Parkinson’s disease.

4. Limitations and Future Work

These findings should be interpreted as a proof-of-concept rather than evidence of clinical readiness. The most important limitation is that the dataset included multiple recordings per person, but the train-test split was performed at the recording level. As a result, recordings from the same individual could potentially appear in both the training and test sets. This could inflate test performance because the model may partly learn person-specific voice characteristics rather than patterns that generalize to completely new patients. Therefore, the reported ROC-AUC, sensitivity, and specificity may be optimistic.

Future work should validate the model on a larger external dataset and use patient-level train-test splitting to better assess performance on unseen individuals. Overall, this analysis suggests that RBF-kernel SVMs can classify Parkinson’s disease status from biomedical voice features with strong performance in this dataset, but further validation is needed before the model can be considered clinically generalizable.

References

1. Little MA, McSharry PE, Hunter EJ, Spielman J, Ramig LO. Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. *IEEE Transactions on Biomedical Engineering*. 2009;56(4):1015–1022. doi:10.1109/TBME.2008.2005954.
2. UCI Machine Learning Repository. Parkinsons Dataset. Irvine, CA: University of California, Irvine; 2008. Accessed June 8, 2026.
3. Cortes C, Vapnik V. Support-vector networks. *Machine Learning*. 1995;20:273–297.
4. Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*. 2011;12:2825–2830.